

Shuanghao Bai

Tel: +86-199-4940-8829 | Email: baishuanghao@stu.xjtu.edu.cn

 [Homepage](#) |  [Google Scholar](#) |  [Github](#)

Xi'an, Shannxi - China

EDUCATION

- **Xi'an Jiaotong University** Sept. 2022 - Jun. 2027
Ph.D. candidate of Artificial Intelligence Xi'an, China
 - Advisor: [Badong Chen](#)
 - Research Interests: Robotics, Vision-Language Models, Generalization,
- **Chongqing University** Sept. 2018 - Jun. 2022
Bachelor of Automation Chongqing, China
 - Advisor: [Min Zhao](#)
 - GPA: 3.68/4.00 (Top 5%)

PROJECTS

- **Cloud-Edge-Device Robot Platform** Sept. 2022 - Dec. 2025
Basic theories and key technologies of cloud-edge-device integrated service robot cloud-brain platform
 - Mainly focuses on generalization tasks in computer vision, addressing challenges posed by limited data and significant distribution shifts between training and test domains. (1) Proposed a simple yet effective method that significantly improves generalization in few-shot settings by incorporating multilayer perceptrons during pretraining [\[C.3\]](#); (2) Unified and evaluated prompt tuning techniques on the vision-language model CLIP for unsupervised domain adaptation, and proposed a prompt-based method to mitigate domain discrepancies [\[C.2\]](#); (3) Developed a generative prompt learning method for domain generalization that leverages CLIP and CGANs to learn and transfer domain-specific prompts to unseen domains [\[C.1\]](#).
- **Robotic Arm Platform (Project Applicant and Leader: Shuanghao Bai)** Jan. 2024 - Dec. 2025
Robotic arm platform technology and application based on visual language action model
 - Developed vision-language-action (VLA) models for robotic manipulation. (1) Analyzed representation redundancy within robotics model using mutual information, and introduced an information bottleneck approach to enhance generalization performance [\[C.5\]](#). (2) Introduced the first VLA model to enable skill chaining in long-horizon tasks [\[C.6\]](#).
- **Multi-agent Collaboration** Jun. 2023 - Dec. 2023
Research on natural human-machine interaction technology for heterogeneous unmanned swarms
 - Designed a system enabling robots (drones and ground fleets) to understand human language and make decisions based on environmental perception. Language understanding is achieved via large language models for task decomposition and code generation. Environmental perception relies on RGB images captured by drones, which are processed by vision-language models to generate heatmaps that guide fleet actions [\[J.1\]](#).

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=IN SUBMISSION

I. Generalization in Vision Language Models

- [C.1] **Shuanghao Bai***, Yuedi Zhang*, Wanqi Zhou, Zhirong Luan, Badong Chen. Soft Prompt Generation for Domain Generalization. In European Conference on Computer Vision (ECCV). 2024. [\[Paper\]](#) [\[Code\]](#)
- [C.2] **Shuanghao Bai**, Min Zhang, Wanqi Zhou, Siteng Huang, Zhirong Luan, Donglin Wang, Badong Chen. Prompt-based Distribution Alignment for Unsupervised Domain Adaptation. In Proceedings of the AAAI Conference on Artificial Intelligence (AAAI). 2024. [\[Paper\]](#) [\[Code\]](#)
- [C.3] **Shuanghao Bai**, Wanqi Zhou, Zhirong Luan, Donglin Wang, Badong Chen. Improving Cross-domain Few-shot Classification with Multilayer Perceptron. In IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). 2024. [\[Paper\]](#) [\[Code\]](#)

- [C.4] Haoran Zhang*, **Shuanghao Bai***, Wanqi Zhou, Jingwen Fu, Badong Chen. PromptTA: Prompt-driven Text Adapter for Source-free Domain Generalization. In IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). 2025. [\[Paper\]](#) [\[Code\]](#)
- [S.1] Wanqi Zhou*, **Shuanghao Bai***, Qibin Zhao, Badong Chen. Revisiting the Adversarial Robustness of Vision Language Models: a Multimodal Perspective. ArXiv preprint arXiv: 2404.19287. [\[Paper\]](#) [\[Code\]](#)
- [S.2] Yuedi Zhang, **Shuanghao Bai**, Wanqi Zhou, Zhirong Luan, Badong Chen. Dual-Path Stable Soft Prompt Generation for Domain Generalization. ArXiv preprint arXiv: 2505.18770. [\[Paper\]](#) [\[Code\]](#)

II. Robot Learning

- [C.5] **Shuanghai Bai**, Wanqi Zhou, Pengxiang Ding, Wei Zhao, Donglin Wang, Badong Chen. Rethinking Latent Representations in Behavior Cloning: An Information Bottleneck Approach for Robot Manipulation. In International Conference on Machine Learning (ICML). 2025. [\[Paper\]](#) [\[Code\]](#) [\[Project\]](#)
- [C.6] Yiguo Fan*, Pengxiang Ding*, **Shuanghao Bai***, Xinyang Tong*, Yuyang Zhu, Hongchao Lu, Fengqi Dai, Wei Zhao, Yang Liu, Siteng Huang, Zhaoxin Fan, Badong Chen, Donglin Wang. Long-VLA: Unleashing Long-Horizon Capability of Vision Language Action Model for Robot Manipulation. In Conference on Robot Learning (CoRL). 2025. [\[Paper\]](#) [\[Project\]](#)
- [C.7] Wei Zhao, Pengxiang Ding, Zhang Min, Zhefei Gong, **Shuanghao Bai**, Han Zhao, Donglin Wang. VLAS: Vision-Language-Action Model with Speech Instructions for Customized Robot Manipulation. In International Conference on Learning Representations (ICLR). 2025. [\[Paper\]](#) [\[Code\]](#)
- [J.1] Zhirong Luan, Yijun Lai, Rundong Huang, **Shuanghao Bai**, Yuedi Zhang, Haoran Zhang, Qian Wang. Enhancing Robot Task Planning and Execution through Multi-Layer Large Language Models. In Sensors. 2024. [\[Paper\]](#)
- [S.3] Can Cui, Pengxiang Ding, Wenxuan Song, **Shuanghao Bai**, Xinyang Tong, Zirui Ge, Runze Suo, Wanqi Zhou, Yang Liu, Bofang Jia, Han Zhao, Siteng Huang, Donglin Wang. Openhelix: A short survey, empirical analysis, and open-source dual-system vla model for robotic manipulation. ArXiv preprint arXiv: 2505.03912. [\[Paper\]](#) [\[Code\]](#) [\[Project\]](#)
- [S.4] **Shuanghao Bai**, Wenxuan Song, Jiayi Chen, Yuheng Ji, Zhide Zhong, Jin Yang, Han Zhao, Wanqi Zhou, Wei Zhao, Zhe Li, Pengxiang Ding, Cheng Chi, Haoang Li, Chang Xu, Xiaolong Zheng, Donglin Wang, Shanghang Zhang, Badong Chen. Towards a Unified Understanding of Robot Manipulation: A Comprehensive Survey. ArXiv preprint arXiv: 2510.10903. [\[Paper\]](#) [\[Code\]](#)
- [S.5] Haoran Zhang*, **Shuanghao Bai***, Wanqi Zhou, Yuedi Zhang, Qi Zhang, Pengxiang Ding, Cheng Chi, Donglin Wang, Badong Chen VCoT-Grasp: Grasp Foundation Models with Visual Chain-of-Thought Reasoning for Language-driven Grasp Generation ArXiv preprint arXiv: 2510.05827. [\[Paper\]](#) [\[Code\]](#) [\[Project\]](#)

III. Causal Learning in Machine Learning

- [C.8] Wanqi Zhou, **Shuanghao Bai**, Shujian Yu, Qibin Zhao, Badong Chen. Jacobian Regularizer-based Neural Granger Causality. In International Conference on Machine Learning (ICML). 2024. [\[Paper\]](#) [\[Code\]](#)
- [J.2] Wanqi Zhou, **Shuanghao Bai**, Yicong He, Badong Chen. An Information-Theoretic Approach for Heterogeneous Differentiable Causal Discovery. In Neural Networks. 2025. [\[Paper\]](#) [\[Code\]](#)

HONORS AND AWARDS

• National Scholarship	Dec. 2024
• National Third Prize in the Phoenix Intelligent Technology Innovation and Application Competition	Jun. 2021
• Grade A Comprehensive Scholarship in Chongqing University	Dec. 2020
• Outstanding Individual Youth Volunteer of Chongqing University	May. 2020
• National Scholarship	Dec. 2019
• Outstanding Student of Chongqing University	Dec. 2019

SKILLS

- **Programming Languages:** Python, Pytorch, C++
- **Languages:** Chinese, English

EXCHANGE AND INTERNSHIP EXPERIENCE

Beijing Academy of Artificial Intelligence (BAAI)	2025.08 – 2025.11
Center for Embodied Multimodal Foundation Models – Research Intern	Advisor: Cheng Chi & Shanghang Zhang
Westlake University	2024.09 – 2025.06
MiLAB – Visiting Student	Advisor: Donglin Wang

ACADEMIC SERVICE

- **Conference Reviewer:** AACL, ICLR, UAI, ICRA, ICIRA
- **Journal Reviewer:** TIP, TCSVT, KBS, NN, Neucom, CAAI TIT